

A review of ‘Fermat’s Enigma’

Kashish Jaswani

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Hanc marginis exiguitas non caperet.

Transl. The smallness of the margin would not contain this.

Pierre De Fermat, one of the greatest mathematicians to have ever lived, seems to have had a flair for mischief. He often taunted his contemporaries, dared them to find proofs of theorems he had already discovered. He did not deem it necessary to write a full proof of any theorem, and merely wrote out his observations in his copy of the *Diophantus’s Arithmetica*. In one such instance, he claimed that he had a ‘truly marvelous proof’ of his statement, but the margins were too small to contain it. Unbeknownst to him, this problem would torment mathematicians for the next three centuries.

Countless limericks surround this problem, each warning the mathematician to stay away from the question which evaded the likes of Euler and Gauss. But by some luck, these did not deter young Andrew Wiles from deciding at the age of just ten, that he would solve this problem. His obsession would eventually lead to one of the most beautiful proofs in mathematics, which not only put an end to the mystery, but also led to insightful and brilliant mathematics. It is this inspiring journey of perseverance, determination and belief in oneself, which Simon Singh has chronicled in his book, ‘Fermat’s Enigma.’

To understand the importance of Wiles’ work, it is imperative to know the history of the problem, and more importantly, how years of work, by many different mathematicians have led to this proof. Simon Singh has managed to convey all this and more, starting all the way from Pythagoras in sixth century BC, to the events that led to the proof in 1994. He has managed to include all major landmarks in mathematical history, and has provided a brief account of almost every mathematical figure mentioned in undergraduate texts.

The first major theorem that schoolchildren are taught is the famous *Pythagoras theorem*. Given a right angled triangle with sides x and y and hypotenuse z , the following equation holds

$$x^2 + y^2 = z^2$$

Conversely, any triangle in which the above equation holds is a right angled triangle. We know some solutions to this equation, (3, 4, 5) and (5, 12, 13). In fact, there are infinitely many such triples, which was proved by the Greeks. Now, a slight modification in the exponent gives us the following equation,

$$x^3 + y^3 = z^3$$

Interestingly, this equation does not have a whole number solution (except the trivial case). Neither does

$$x^4 + y^4 = z^4.$$

Fermat stated that, in general,

$$x^n + y^n = z^n$$

has no whole number solutions for $n \geq 2$. This is known as *Fermat's Last Theorem*.

These types of questions come from an area of mathematics known as Number Theory. This subject involves study of natural numbers and particularly, relationships between them. Akin to other scientific disciplines, the more you try to understand at the atomic level, the more mysterious it tends to get. In mathematics, prime numbers serve as building blocks, and there are many unanswered questions about the behavior of these numbers. Number theory, amongst other things, seeks to provide insight into the behavior of primes. An example of an unsolved problem is the *Twin Prime Conjecture*. This asks whether there are infinitely many pairs of primes which are two numbers apart, such as 3 and 5, or 11 and 13. Often problems in number theory are deceptively simple. They are generally easy to understand, but notoriously difficult to solve. Thus, we often employ tools from analysis and geometry in number theoretic proofs.

The first chapter of Fermat's Enigma focuses on the Pythagorean Brotherhood and their work in Number Theory. This school was established by Pythagoras in Croton with the aim of understanding mathematical techniques used by the Egyptians and the Babylonians. Despite being very secretive, their ideas spread and were built upon by many mathematicians. Today several interesting concepts (and unsolved questions), such as those of perfect numbers and slightly excessive numbers originated from their work. Pythagoras was also interested in understanding the fundamental role mathematical laws play in nature. Their work in music theory and planetary motion, led them to declare 'Everything is Number!' They believed that they were developing a language to 'describe the nature of the universe.'

Pythagoras' death led to a series of events at the end of which city of Alexandria became the mathematical hub of ancient Egypt. The famous Alexandrian Library housed books containing all existing mathematical knowledge in the world, and the person in charge was Euclid. The author then details Euclid's contribution to mathematics (such as *irrational numbers*, *proof by contradiction* and *Euclid's Elements*). Simon Singh then takes us through time, through Diophantus' life and death, and eventually the burning of Alexandria which led to the beginning of the Dark Ages. At the end of these tumultuous times, books from Alexandria which had miraculously survived had now reached other parts of the world.

One such work was Diophantus's *Arithmetica* which had found its way to Pierre De Fermat, who breathed new life into the subject of number theory. Fermat, a recluse, only solved problems when he was not busy being a judge. He had a peculiar way of studying the subject - he did not bother writing any of his proofs formally, and as soon as he convinced himself of an answer, he would move on to the next problem. He did not communicate with his fellow mathematicians, except when he was busy founding probability theory with Blaise Pascal. However his observations are astounding, and he discovered deep patterns in numbers, many of which have taken mathematicians a considerably long amount of time to prove again. One such example is the fact that 26 is the only number which is in between a square number ($25 = 5^2$) and a cube number ($27 = 3^3$).

After his death, Fermat's son recognised his fathers genius and decided to rewrite his observations and publish an annotated version of *Arithmetica*. These ideas spread and mathematicians

all around the world became busy trying to find proofs amongst his hasty scribbles. They were successful in all but one case. In fact, it is called Fermat's *Last* Theorem, because it was the only one which remained unproven.

Fermat gave a brief description of how to go about showing that $x^n + y^n = z^n$ has no solutions when $n = 4$. Euler then formalised this method known as *infinite descent*. Euler eventually modified the proof for the case when $n = 3$, but could not develop his idea for the general case. Then, the French mathematician Sophie Germain showed that no solutions existed for primes of the form p , such that $2p + 1$ is also a prime (known as Germain primes). Further, this method was adapted by Dirichlet and Legendre for when $n = 5$ and Gabriel Lamé for $n = 7$. Attempts of the general proof however remained largely unsuccessful. Years of effort had only generated a proof for a few special cases. This formidable beast had left even the best of mathematicians bewildered. As we were approaching the twentieth century, it was becoming increasingly clear that known techniques were not enough to find a general proof.

Around 1955, Japanese collaborators Goro Shimura and Yutaka Taniyama came up with an exceptional idea to link two seemingly unrelated concepts of mathematics, elliptic curves and modular forms. An elliptic curve is an equation of the form

$$y^2 = x^3 + ax + b$$

where a, b are fixed whole numbers. A natural way to understand the behaviour of these equations is by trying to find their integer solutions. However, this is difficult for Diophantine equations in general (polynomial equations for which we only seek integer solutions). While mathematicians use techniques such as infinite descent, going modulo or even brute force, there are many equations which evade all these. One of Hilbert's twenty-three problems states this formally,

Given a Diophantine equation with any number of unknown quantities and with rational integral numerical coefficients: To devise a process according to which it can be determined in a finite number of operations whether the equation is solvable in rational integers.

Yuri Matiyasevich proved in 1970 that such an algorithm does not exist. This problem continues to remain unsolved if integers are replaced with rationals.

To make the job easier, mathematicians instead look for solutions to elliptic curves in a finite field $\mathbb{F}_p$. Based on the number of solutions for each p , we can also associate an L-series with each elliptic curve.

Modular forms, on the other hand, have their origins in the subject of analysis. They are also useful in subjects such as algebraic topology, string theory and cryptography. These special type of complex functions have a truly marvelous property - they satisfy infinitely many symmetries! In fact, this is so unusual, that it is a surprise that they exist at all.

Let $\mathbb{H} = \{z \in \mathbb{C} | \text{Im}(z) > 0\}$. Define a modular form to be a function

$$f : \mathbb{H} \rightarrow \mathbb{C}$$

such that

1. f is holomorphic
2. $f\left(\frac{az+b}{cz+d}\right) = f(z)(cz+d)^k$ where $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$ and k depends on the modular form
3. $f(z) = \sum_{n=0}^{\infty} a_n e^{2\pi i n z}$

Using the coefficients a_n , we can associate a M-series with every modular form.

Since these functions lie in dimension four, it is difficult to visualise them. Frank Farris proposed the method of *domain colouring* which can be used to plot such functions. Here we assign a color and saturation level to every point on the complex plane. Then we view the phase portrait, where every point $z \in \mathbb{C}$ has the color associated with $f(z)$. We can use this method to visualise modular forms and observe their symmetry. Their symmetry is useful because to understand the function, we only need to know how the function behaves on a *fundamental domain*. Due to symmetry, it will behave the same everywhere else (condition 2 above guarantees symmetry).

During his work, Taniyama came across a modular form, whose associated M-series matched the L-series of an elliptic equation. This seemed to hold for all subsequent modular forms he analysed. Over time, along with Shimura, their intuition led them to an unbelievable conclusion - that every elliptic curve has a modular form and vice versa.

Since this was so unexpected, the community of mathematicians needed some convincing and years of experimentation by the Japanese duo had started to convert skeptics around the world. In Fermat's Enigma, the development of this conjecture has been described in immense detail. The author does justice to these mathematicians by providing the full account of Taniyama's discovery, and the belief Shimura had on this idea which he dedicated all his time to proving after Taniyama's death. The style of writing does not only focus on one particular event, rather, the author has not shied away from taking all the necessary tangents to provide the overall mathematics culture in the time period he is mentioning. Hence, this chapter digresses into the repercussions of the conjecture and particularly the Langlands Programme. Simon Singh accomplishes his detailed delivery through interviews with the mathematicians we speak of, through straightforward explanations of math concepts, and interesting historical tidbits which are not otherwise well known.

One moment worthy of note was when Shimura describes why he believed his inconceivable conjecture to be correct - "I have this philosophy of goodness (...) Mathematics should contain goodness. Most mathematicians do mathematics from an aesthetic point of view and that philosophy of goodness comes from my aesthetic viewpoint."

It is quite unexpected that this conjecture had anything to do with Fermat's Last Theorem. But, the rich mathematics of modular forms was as Singh puts it, 'The Missing Link' that was needed. In 1984, Gerhard Frey showed that a proof of this conjecture would instantly prove the Last Theorem. He adopted the method of proof by contradiction and showed that, if the theorem was false, it would lead to such a weird elliptic equation, that it could not be associated with a modular form. This would in turn contradict the conjecture. This was further formalised by Ken Ribet in 1986.

Over time, mathematicians had started to suspect that a proof of Fermat's theorem would not provide any new insight into the theory of numbers and many did not want to spend their entire career pursuing such an elusive proof. But now this theorem was linked to the modularity conjecture which was sure to have great repercussions. In fact, many assumed the conjecture to be true and started proving other results. Andrew Wiles now considered his dream to be a 'professionally acceptable' problem, and he was confident that even if he did not find the solution, he would be doing worthwhile mathematics. Thus began his eight year long period of self enforced solitude.

Wiles spent the first eighteen months just familiarizing himself with the problem. Having ex-

tensively studied elliptic curves during his time at Cambridge, he was aware of almost all major techniques he could start using for this proof. Eventually, he decided to use the method of proof by induction, and used Galois representations to establish the base case. Essentially, he was trying to show that the first element of the L-series of an elliptic curve matches the first element of the M-series of a modular form. Contrary to previous approaches, he did not try to prove this for individual elliptic curves (and find a corresponding modular form), his approach was much more general.

At this stage, two years had passed and Wiles could not think of any new ideas to continue with his proof after the base case. To his relief, Joan Coates, Andrew's PhD supervisor came to his rescue. Coates' student Mattheus Flach was working on elliptic curves, particularly Kolyvagin's method. Andrew spent the next few months studying the Kolyvagin-Flach method through which he could separate elliptic curves into different families, and prove the theorem for each individual family. Eventually, he realised that he had stumbled into the unfamiliar territory of arithmetic geometry, and decided to recruit his colleague Nick Katz. After discovering the final piece of the puzzle in Barry Mazur's paper, he finished the proof in May of 1993.

Simon Singh had the opportunity to interview Andrew Wiles and has included parts of the conversation in his book. These provide us with inspiring insight into Wiles's thought process during the course of working on the proof. Through quotes, we understand why Wiles decided to work alone and what kept him inspired enough to keep going. Perhaps the most strength and resilience was shown by Wiles when, during review, Nick Katz discovered a gap in his proof. Wiles and his former student Richard Taylor, spent another year, but this time with the added pressure of spectators and with countless opinions about his work. Despite the doubt, Wiles persisted and the correct and complete proof was published in 1994. When describing how he felt during the final moments of this long journey, he said, "It was so indescribably beautiful; it was so simple and so elegant. (...) I couldn't contain myself, I was so excited. It was the most important moment of my working life. Nothing I ever do again will mean as much."

The final few chapters of Singh's book talk about these eight years in incredible detail. He has also included emails between mathematicians which reveal the buzz and excitement felt by the community during that time. The author has also managed to convey the revolutionary nature of the proof to the reader. When we look at the final proof, it is a euphony of mathematical ideas that span centuries. Not only are many landmark theorems of number theory used, but it has brought forth ideas which others will build upon in years to come. It is commendable that Singh has gone through great effort to include many of these opinions in his book.

The Taniyama-Shimura conjecture and its proof form a bridge between two areas of mathematics, namely number theory and harmonic analysis. *Do more such bridges exist?* This was a question first asked by Robert Langlands in his letter to number theorist Andre Weil in 1967. This letter contained several conjectures which seemed to link Galois representations and automorphic forms, built on earlier work by Harish Chandra, Israel Gelfand and Emil Artin. Even now, only specific cases on this conjecture have been proved. Wiles' proof served as the first major breakthrough in this project. But, the amount of time it took for the proof, and in proving its generalisations, is a warning of just how difficult it will be to build more such connections.

In the past several years, there have been major advances in the project. In 2024, Dennis Gaitsgory, Sam Raskin and several others published an eight hundred page proof of the Geometric Langlands conjecture which took them thirty years to develop. Such theorems give us a new approach towards

problems which, till now, seemed unsolvable. Now, we could potentially translate them to a different area with known techniques which unlock new possibilities. This venture has laid down the foundations for the next few decades of research in Mathematics. It has the ability to span across subjects like Algebra, Arithmetic Geometry, Galois Theory and Mathematical Physics. Today, many mathematicians across the world are actively working towards building this ‘grand unified theory of Mathematics.’

The book Fermat’s Enigma is a unique retelling of all the ideas involved in solving the Last Theorem. The author has managed to maintain a balance between being technical and yet conveying the importance of the story so that a general audience is able to appreciate it. In fact, any reader would walk away with knowledge about centuries of work in mathematics, and the major figures who helped shape the ideas we study today. Instead of monotonous retellings of these famous mathematicians, Singh has portrayed their stories with an almost theatrical writing style bound to attract any reader. It is extremely well researched, and correspondence between mathematicians have been included to provide a break from the niche details. A lot of ground has been covered, both historically and mathematically and anyone who reads this book is certain to find at least a few topics which they want to explore further.

While the majority of the book is about the Last Theorem, the author realises that this book will be read by many who have not formally studied the subject. Keeping this responsibility in mind, he tries to present the essence of mathematics to this general audience. He has done this through broad discussions surrounding the *importance of proof*, *what motivates a mathematician* and *what constitutes good mathematics*. The book has answered these through Andrew Wiles’ perspective, but there are many debates on such philosophical questions. Any aspiring mathematician will eventually have to find their own answers as they progress on their own academic journey.

References

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